QQ-A-250/28A
September 14, 1973
SUPERSEDING
Int. Fed. Spec. QQ-A-00250/28(ASG)
November 10, 1969

FEDERAL SPECIFICATION

ALUMINUM ALLOY, 7011 ALCLAD 7178, PLATE AND SHEET

This specification was approved by the Commissioner, Federal Supply Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the specific requirements for aluminum alloy, 7011 alclad 7178, plate and sheet; the general requirements are covered in QQ-A-250. The plate and sheet covered by this specification shall be an integral composite product consisting of a heat-treatable aluminum alloy (7178) core with thin layers of an aluminum alloy (7011) anodic to the core and of approximately equal thickness, bonded to both surfaces.

1.2 Classification.

1.2.1 Tempers. The plate and sheet shall be classified as O, T6, T62, T651, T76, T7651, or F temper, as specified (see 6.2). The definition of these tempers shall be as specified in ANSI H35.1 or as follows:

O	- Annealed.
T6	- Solution-heat-treated and artificially aged.
T62	- Solution-heat-treated from the O or F temper to demonstrate the response to heat-treatment, and artificially aged.
T651	- Solution-heat-treated, stress relieved by stretching to produce a permanent set of 2 percent nominal but not less than 1-1/2 nor more than 3 percent, and artificially aged. Plate shall receive no further straightening after stretching.
T76	- Solution-heat-treated and artificially aged sufficient to produce improved resistance to exfoliation.
T7651	- Solution-heat-treated, stress relieved by stretching to produce a permanent set of 2 percent nominal but not less than 1-1/2 nor more than 3 percent, and artificially aged sufficient to produce improved resistance to exfoliation and stress corrosion cracking. Plate shall receive no further straightening after stretching.
F	- As fabricated.

FSC 9535

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

QQ-A-250	Aluminum Alloy Plate and Sheet; General Specification for

Federal Standards:

FED-STD-184	Identification Marking of Aluminum, Magnesium and Titanium

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-I-8950 - Inspection, Ultrasonic, Wrought Metals, Process for.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

ASTM D 1193	Standard Specification for Reagent Water
ASTM STP425 (1967)	Stress Corrosion Testing (Report of ASTM TG-1 Committee B-3)

(Application for copies of ASTM standard should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

American National Standards Institute

ANSI H35.1	Alloy and Temper Designation System for Aluminum

(Application for copies of ANSI standard should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Chemical composition.

3.1.1 The chemical composition of the core ingots or slabs and of the cladding plates used for the manufacture of the clad plates and sheets shall conform to the requirements shown in table I for core and cladding, respectively.

TABLE I. Chemical Composition 1/

Element	Analysis			
	Core alloy (7178) Percent		Cladding alloy (7011) Percent	
	Minimum	Maximum	Minimum	Maximum
Zinc	6.3	7.3	4.0	5.5
Magnesium	2.4	3.1	1.0	1.6
Copper	1.6	2.4	--	0.05
Chromium	0.18	0.35	0.05	0.20
Manganese	--	0.30	0.10	0.30
Iron	--	0.50	--	0.20
Silicon	--	0.40	--	0.15

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TABLE I. Chemical Composition 1/ (con.)

Element	Analysis			
	Core alloy (7178) Percent		Cladding alloy (7011) Percent	
	Minimum	Maximum	Minimum	Maximum
Titanium	--	0.20	--	0.05
Others, each	--	0.05	--	0.05
Others, total	--	0.15	--	0.15
Aluminum	Remainder		Remainder	

1/ Analysis shall regularly be made only for the elements specifically mentioned in table I. If, however, the presence of other elements is indicated in the course of routine analysis, further analysis shall be made to determine conformance to the limits specified for other elements.

3.2 Mechanical properties.

3.2.1 The mechanical properties of materials as supplied. The mechanical properties perpendicular to the direction of final rolling, except for material less than 9 inches in width, shall conform to the requirements of table II for the temper specified. For material less than 9 inches in width, the mechanical properties parallel to the direction of final rolling shall conform to the requirements of table II for the temper specified.

3.2.2 Mechanical properties, after heat treatment. In addition to conforming to the requirements of 3.2.1, material in the annealed (O) and the as-fabricated (F) tempers, after solution heat treatment and aging, shall also conform to the requirements of table II for the solution-heat-treated and artificially aged temper (T6). Material ordered in the annealed (O) temper and the as-fabricated (F) temper, after solution heat treatment and stabilizing, shall be capable of conforming to the requirements specified in table II for the T76 temper.

3.2.3 Bend test. Bend-test specimens taken from plate and sheet shall be capable of withstanding, without cracking, the bend test specified in QQ-A-250. The values for bend factor N are given in table III.

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TABLE II. Mechanical Properties

Temper	Thickness in inches	Tensile Strength (psi) minimum	Yield strength at 0.2 percent offset (psi) minimum	Elongation in 2 in. or 4D (percent) minimum 1/ 2/
O	0.015 thru 0.499	40,000 3/	21,000 3/	10
	0.500 thru 2.000	40,000 3/ 4/	--	10
T6 and T62 5/	0.015 thru 0.044	79,000	69,000	7
	0.045 thru 0.062	81,000	70,000	8
	0.063 thru 0.187	82,000	71,000	8
	0.188 thru 0.249	83,000	72,000	8
T651 and T62 5/	0.250 thru 0.499	83,000	72,000	8
	0.500 thru 1.000	84,000 4/	73,000 4/	6
	1.001 thru 1.500	84,000 4/	73,000 4/	4
	1.501 thru 2.000	80,000 4/	70,000 4/	3
T76	0.045 thru 0.062	73,000	62,000	8
	0.063 thru 0.187	73,000	62,000	8
	0.188 thru 0.249	74,000	63,000	8
T7651	0.250 thru 0.499	73,000	61,000	8
	0.500 thru 1.000	73,000 4/	62,000 4/	6
F	All	6/	6/	6/

1/ Not required for material 1/2 inch or less in width.
2/ D represents specimen diameter.
3/ Maximum.
4/ The properties for these thicknesses are those of the core alloy since the tests are made on a round specimen machined from the plate.
5/ Plate in the T62 temper is not available from the material producer; the properties listed indicate those which can usually be obtained by the user, T62 condition, when the material is properly solution- and precipitation-heat-treated from the O (annealed) of F (as fabricated) temper. These properties also apply to samples of material supplied in the O or F tempers, which are solution- and precipitation-heat-treated by the producer to determine that the material will respond to proper heat treatment. Properties attained by the user, however, may be lower than those listed if the material has been formed or otherwise cold- or hot-worked, particularly in the annealed temper, prior to solution heat treatment.
6/ No requirements.

TABLE III. Bend Test Factor "N"

Thickness in inches	Tempers	
	O-annealed	T6, T62 & T76
0.015 thru 0.020	1	7
.021 thru .062	2	8
.063 thru .091	3	9
.092 thru .125	4	10
.126 thru .249	5	11
.250 thru .499	6	14

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3.3 Cladding thickness.

3.3.1 Thickness of cladding plates. The aluminum alloy (7011) plates that are bonded to the two sides of the aluminum alloy (7178) ingot or slab to form a composite that is to be rolled to the finished thickness shall each have a thickness as specified in table IV.

TABLE IV. Cladding Thickness

Thickness of finished plate or sheet in inches	Nominal cladding thickness per side; percent of composite thickness	Average cladding thickness per side on finished plate or sheet; min. percent of plate or sheet thickness
Up thru 0.062	4.0	3.2
0.063 thru 0.187	2.5	2.0
0.188 and over 1/	1.5	1.2

1/ For plate 0.500 inch and over, the average cladding thickness per side shall have a maximum value of 3 percent of the plate thickness.

3.3.2 Thickness of cladding. If question arises concerning the thickness of cladding of the finished sheet or plate, samples examined in accordance with QQ-A-250 shall show an average of cladding on each side not less than that specified in table IV.

3.4 Internal defects. When specified (see 6.2), plate shall be ultrasonically inspected (see QQ-A-250). Acceptance limits shall be as specified in table V.

TABLE V. Ultrasonic Discontinuity Acceptance Limits 1/

Thickness in inches	Maximum weight per piece in pounds	Discontinuity class 2/
0.500 thru 1.499	2,000	B
1.500 thru 2.000	2,000	A

1/ Discontinuities in excess of those listed in table V may be allowed, subject to the approval of the procuring activity, if it is established that they will be removed by machining or that they are in non-critical areas.

2/ The discontinuity class limits are defined in MIL-I-8950.

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3.5 Exfoliation and stress-corrosion cracking. Material in T76 and the T7651 tempers, when tested in accordance with 4.2.2 and 4.2.3, shall show no exfoliation or stress corrosion cracking susceptibility beyond the acceptance criteria as detailed in 4.2.2.7 and 4.2.3.5.

3.5.1 Lot acceptance control criteria for T76 and T7651 tempers. Susceptibility to exfoliation and stress-corrosion cracking for each lot of T76 and T7651 temper material shall be determined by the following control criteria (see 4.2.1.2 and 4.2.4).

a. Determine mechanical properties and electrical conductivity.

b. If the sub-surface conductivity is 38 percent IACS or higher, and the tensile properties meet the minimum limits specified herein, the material is acceptable.

c. If the sub-surface conductivity is at least 35 percent IACS but less than 38 percent IACS, the material shall be tested as specified in 4.2.2, or reprocessed.

d. If the sub-surface conductivity is below 35 percent IACS, the material is not acceptable and must be reprocessed.

3.6 Marking. In addition to the marking required in Fed. Std. No. 184, plate and sheet in the T6, T76, T651 and T7651 tempers shall be identified by a lot number marked in at least one location on each piece.

4. QUALITY ASSURANCE PROVISIONS

(See QQ-A-250).

4.1 Mechanical tests after heat treatment.

4.1.1 Number of tests after heat treatment. From material in the annealed (O) and as-fabricated (F) tempers, an additional number of specimens equal to that required by QQ-A-250 shall be taken and tested after solution heat treatment and artificial aging to determine compliance with 3.2.2.

4.2 Exfoliation and stress-corrosion cracking tests (T76 and T7651 only).

4.2.1 Sampling.

4.2.1.1 For exfoliation and stress-corrosion cracking tests. Two samples shall be taken for each 4,000 pounds or less of the first three production lots for each

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size range of T76 and T7651 temper sheet and plate listed in table II. Thereafter, surveillance testing shall be performed on at least one sample per month for each size range of sheet and plate (produced during the month) which had previously been produced with acceptable exfoliation resistance. The surveillance test samples shall be taken from a lot which has met the requirement of 3.5.1 b. Samples as above shall be taken for the stress-corrosion cracking test when thickness permits (see 4.2.3).

4.2.1.2 Sampling for conductivity tests. Sampling shall be in accordance with QQ-A-250 for mechanical property tests. Mechanical properties and conductivities shall be determined on the same samples except cladding must be removed before testing for conductivity.

4.2.1.3 Records. The producer shall maintain records of the performance of all lots sampled and tested in accordance with 4.2.1.1. Upon request of the procuring activity, such records shall be made available. These records shall be maintained for not less than one year after completion of the contract or purchase order.

4.2.2 Resistance to exfoliation and stress-corrosion cracking. Tests shall be performed in accordance with the following procedures.

4.2.2.1 Specimens for exfoliation test. Full thickness specimens of sheet and plate large enough to provide visual assessment of attack may be used. A minimum panel size of 2 inches by 4 inches shall be used. Sawed edges need not be machined smooth; however, panels obtained by blanking or shearing shall have edges machined to a depth equal to the thickness of material to remove cold worked metal. For all materials, 0.100 inch or more in thickness, 10 percent of the thickness shall be removed by machining one surface. The machined surface shall be evaluated by exposure to the test solution. Protection by masking of edges and back surface is not required provided compliance with 4.2.2.4 is assured. Specimens shall be thoroughly degreased prior to testing; however, etching or other pretreatment is not necessary.

4.2.2.2 Equipment. The exfoliation test shall be performed in a suitable inert container of glass or plastic, suitable to be used for the solution and the specimens during the period of exposure. The container should be equipped with a cover to reduce evaporation.

4.2.2.3 Corrosive solution. The corrosive solution shall be prepared by dissolving the following reagent grade chemicals in distilled or deionized water, conforming to ASTM D-1193, Type II, in the indicated concentrations as follows:

Sodium Chloride (NaCl)	234 g/liter of solution
Potassium Nitrate (KNO_3)	50 g/liter of solution
Nitric Acid (HNO_3) (concentrated-70% by weight)	6.3 ml/liter of solution

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The apparent pH of the solution is 0.4. The solution shall be maintained at a temperature of 77 ±5° F (25 ± 3° C) during the test. At the start of each test, a fresh solution shall be used.

4.2.2.4 Exposure procedure. The specimens shall be immersed in a volume of solution sufficient to provide a minimum volume to a specimen surface area ratio of 50 ml per square inch. Unless the edges and the back of the specimen are masked, their area must be included in the determination of the total surface area. The specimens shall be immersed in the solution using rods or racks of glass, plastic, or other inert material to support the specimens above the bottom of the container. The specimens should be exposed in a horizontal position in order to prevent loss of corrosion products from the surface of the specimens.

4.2.2.5 Exposure period. The specimens shall be immersed in the solution for a period of 48 hours without change of solution. Inspection of specimens (without cleaning) at intermediate periods may be performed to detect early development of exfoliation. The test may be terminated whenever exfoliation becomes evident.

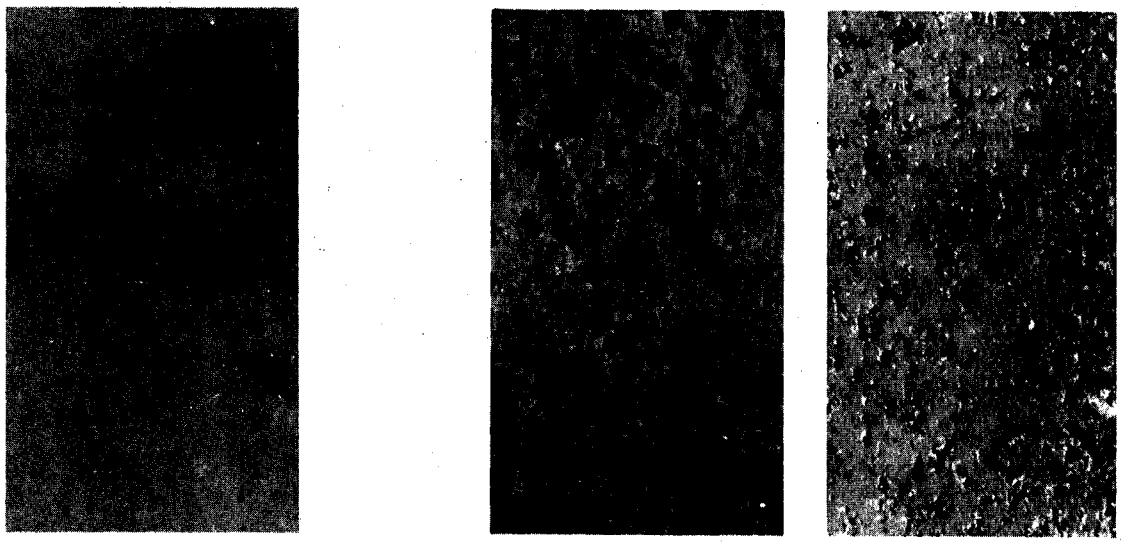
4.2.2.6 Specimen cleaning. At the conclusion of the 48 hour exposure, specimens shall be removed from the solution, gently rinsed in a container of still water, and then cleaned by immersion in concentrated nitric acid at room temperature for approximately 30 minutes, followed by a water rinse and air drying.

4.2.2.7 Test results. Specimens as exposed above shall be visually examined. Surfaces will be either generally roughened or show numerous discrete pits, but shall not show any exfoliation or actual delamination. (Exfoliation as used herein is defined as that form of corrosion which proceeds laterally from the sites of initiation along planes, generally grain boundaries, parallel to the surface. The resulting corrosion products force the metal upwards resulting in a layered appearance to the surface.) In some cases, the pitting attack will have a tendency to undermine the surface, resulting in tiny slivers of metal protruding around the edges of pits. This form of attack does not progressively worsen with continued exposure and should not be misconstrued as exfoliation. Examples of surfaces which have completed this test without exfoliation are shown in figure 1, and surfaces showing various degrees of exfoliation failures are shown in figure 2.

4.2.3 Stress-corrosion cracking test. Stress corrosion cracking tests on plate 0.750 inch and over in thickness shall be performed as follows:

4.2.3.1 Specimens. Specimens shall be selected in a manner such as to permit application of the specific tension stress in the short transverse direction. Standard "c" rings are preferred. (See Report of ASTM TG-1, Committee B3; ASTM-STP425, 1967, Stress Corrosion Testing.)

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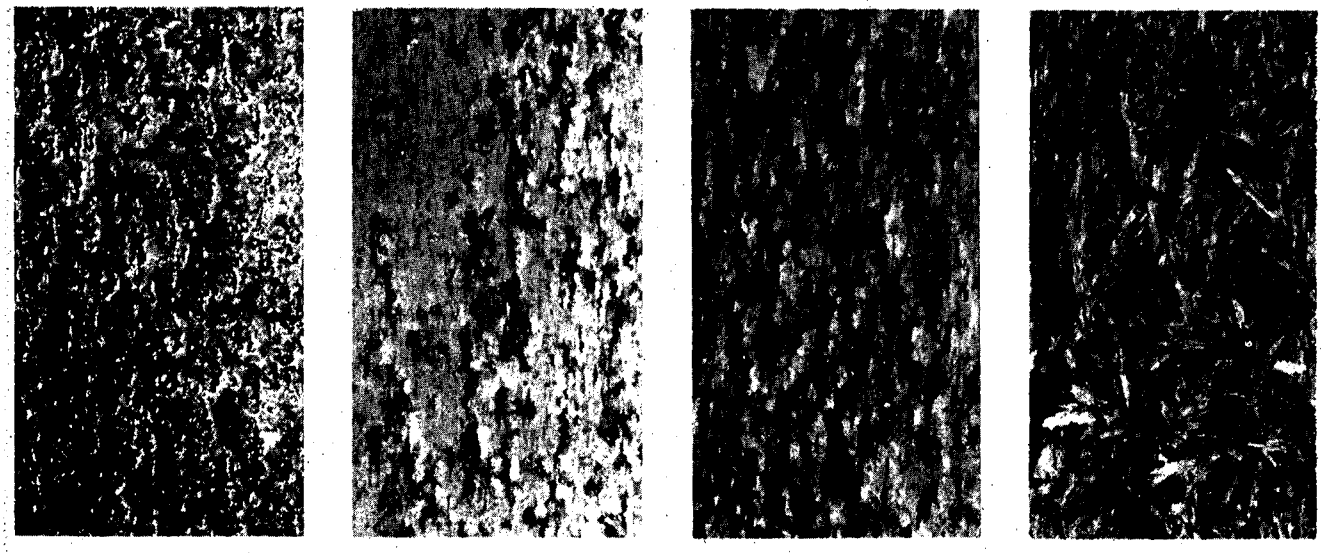


A. No Appreciable Corrosion

B. Range of Attack Encompassed by Pitting Classification

FIGURE I

SURFACE APPEARANCE OF SPECIMENS
NOT SUSCEPTIBLE TO EXFOLIATION



A B C D

FIGURE II

FOUR DEGREES OF SEVERITY OF EXFOLIATION CORROSION

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4.2.3.2 Exposure period. Exposure shall be for 30 days.

4.2.3.3 Tension stress. The tensile stress applied in the short transverse direction (perpendicular to grain flow) shall be 25,000 psi, and the specimen shall be held at constant strain.

4.2.3.4 Exposure procedure. The stressed specimens shall be exposed to a solution of 3.5 percent NaCl prepared with distilled water and reagent grade salt, pH 6.4 to 7.2 at a solution temperature of 75 ±2° F, air temperature 80 ±2° F and air relative humidity of 45 ±6 percent. The alternate immersion cycle shall consist of 10 minutes immersion in solution and 50 minutes in air, with sufficient circulation to dry the specimens slowly before the next immersion.

4.2.3.5 Stress-corrosion test results. After a 30-day exposure, as described in 4.2.3.4, the specimen shall exhibit no evidence of stress-corrosion cracking. Any highly directional attack which is suspected of concealing a stress-corrosion crack shall be cross sectioned and examined metallographically. An example of stress corrosion cracking is shown on figure 3. Figure 4 illustrates pitting type attack which does not constitute failure.

4.2.4 Electrical conductivity tests. The conductivity shall be determined by taking three electrical conductivity readings at random on the machined surface after machining one surface of the sample to a depth of approximately 10 percent of the product thickness. The average of each of the three readings shall be used as the conductivity acceptance criteria in 3.5.1.

5. PREPARATION FOR DELIVERY

(See QQ-A-250)

6. NOTES

6.1 Intended use. 7011 Alclad 7178 plates and sheets are used where fatigue strength and mechanical properties higher than those provided by 7072 Alclad 7178 are required.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Form and temper required (see 1.2.1).
c. Dimensions required.
d. Requirements for sizes not specificially covered (see QQ-A-250).
e. Whether ultrasonic inspection of material is required (see 3.4).
f. Selection of applicable levels of preservation, packaging, and packing required, if other than level C (see QQ-A-250).

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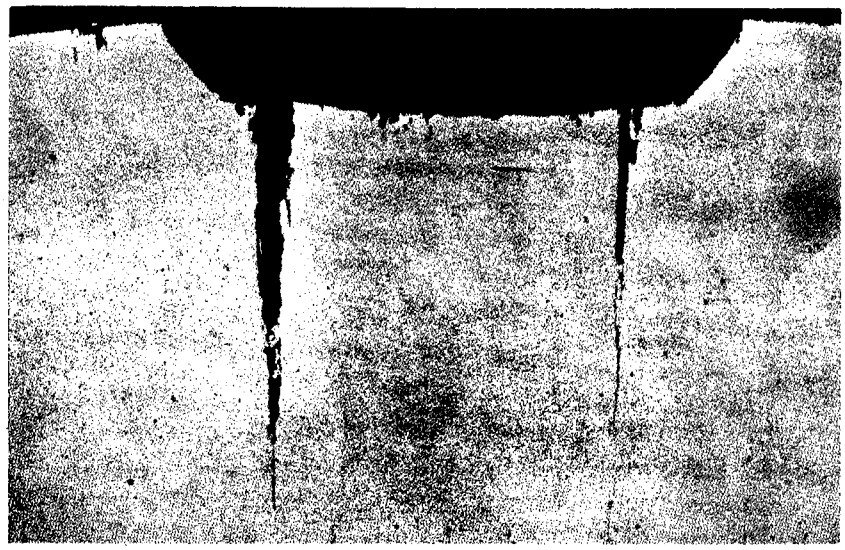


FIGURE 3.

75X - UNETCHED, STRESS CORROSION CRACKS IN SPECIMEN, NOT ACCEPTABLE

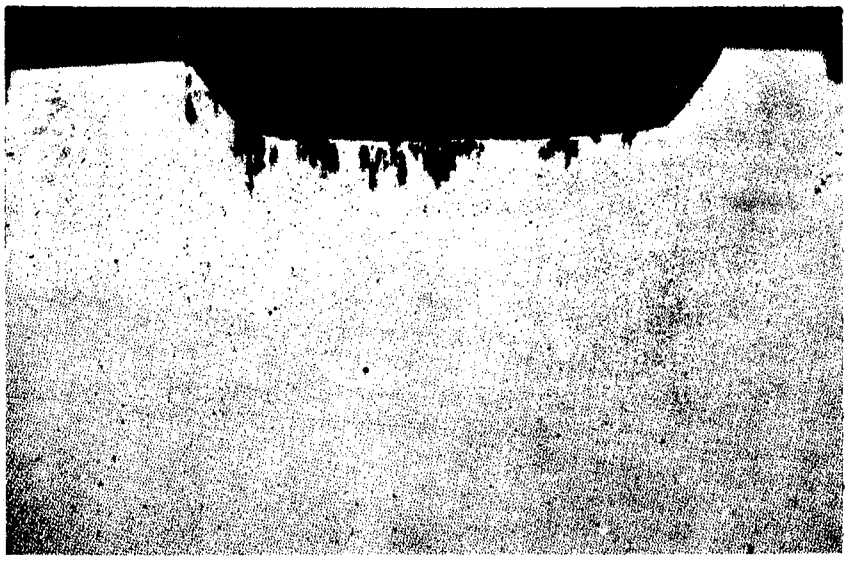


FIGURE 4.

75X - UNETCHED, PITTING TYPE ATTACK, ACCEPTABLE

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6.3 Temper. Tempers T651 and T7651 are available in plate only. Tempers T6 and T76 are available in sheet only.

6.4 Definitions.

6.4.1 Capable of. The term "capable of" as used in the detailed specification means that the test need not be performed by the producer of the plate and sheet. However, should subsequent testing by the procuring activity establish that the material does not meet these requirements, the plate and sheet are subject to rejection.

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DOCUMENT IDENTIFIER AND TITLE
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NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

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